

V3 empirical study · Instruction

Seven models · four underlyings · four volatility regimes · 1.5-year fixed calibration

What this report is

- Computational source of truth: the companion Jupyter notebook. This PDF is written from the same payload; every table number is identical.
- Question: which of seven models tracks realized stock price S_t most closely, and does the ranking change across companies and the four volatility regimes?
- This report does not price options. Option RMSE lives in the companion decision PDF.
- In every table the BEST row is the lowest percentage RMSE of the p50 path versus realized S_t . That row is bold and shaded green; Mark ranks 1–7.

Experimental design (held fixed for every model)

- Models: GBM, Modified GBM (Markov direction), GARCH(1,1), Heston (no jumps), Merton jump-diffusion, GARCH–Merton, Heston–Merton (Bates).
- Companies: SPY, AAPL, MSFT, AMZN. Each name is calibrated and simulated separately.
- Regimes: Crisis 2008-08-01→2009-07-31; Normal 2014-01-01→2014-12-31; Late-cycle 2018-10-01→2019-09-30; COVID 2019-09-01→2020-08-31.
- Calibration window: 1.5-year ending at the first session of each 1-year evaluation window. Rolling: none (one estimate, held for the whole window). Parameters estimated at t_0 are used only after t_0 .
- P-measure: keep the lookback drift μ . Do not replace $\mu \rightarrow r$ (that replacement is only for LSM option pricing).

V3 empirical study · Instruction (continued)

Seven models · four underlyings · four volatility regimes · 1.5-year fixed calibration

Path construction

- One rolling path cloud per ticker × regime × model, $n_paths = 10000$, seed 42.
- Clock: daily trading days from the first bar of the regime through the regime end. $\Delta t = 1/252$.
- Start at the observed open S_0 . Hold the single t_0 calibration for every day of the regime.
- At each day, summarize the cloud by p10, p25, p50, p75, p90. The reported path is p50, not the Monte Carlo mean.

Metrics

- Primary: $RMSE\% = 100 \times \sqrt{\text{mean}(((p50 - S_t)/S_t)^2)}$. Ranking uses this only.
- $MAE = \text{mean} |p50 - S_t|$. Bias \$ = $\text{mean}(p50 - S_t)$. Bias% = $100 \times \text{mean}((p50 - S_t)/S_t)$.
- ICP 25–75 = share of S_t inside $[p25, p75]$. ABW = $\text{mean}(p75 - p25)$. ICP 10–90 is recorded but not a table column.
- ICP is higher-better and is not used for the green row.

Page 3 · Table block · SPY · seven models × four regimes

P-measure daily cloud, reported path = p50. Bold green = lowest RMSE% vs realized S_t. 1.5-year lookback, rolling=none, n_paths=10000, seed=42.

Table SPY-1 SPY · 2008-2009 Crisis
2008-08-01 → 2009-07-31 · n = 252 days

Model	RMSE%	MAE	Bias \$	Bias%	ICP 25-75	ABW	Mark
GBM	34.41	25.9924	+25.4098	+29.70	11.1%	19.6746	2
Modified GBM	35.81	27.1703	+26.6664	+31.11	9.9%	18.2026	3
GARCH	39.87	30.5643	+30.0894	+35.01	11.9%	22.4196	7
Heston	37.73	28.7475	+28.3468	+32.99	10.7%	21.6058	5
Merton	30.85	22.9129	+22.2314	+26.10	14.7%	19.0387	1
GARCH-Merton	39.77	30.4684	+30.0004	+34.91	11.9%	22.8520	6
Heston-Merton	37.07	28.1515	+27.7450	+32.32	11.5%	21.9117	4

Table SPY-2 SPY · 2013-2014 Normal
2014-01-01 → 2014-12-31 · n = 253 days

Model	RMSE%	MAE	Bias \$	Bias%	ICP 25-75	ABW	Mark
GBM	5.00	8.1840	+8.1743	+4.20	81.4%	21.1529	1
Modified GBM	5.34	8.8286	+8.8206	+4.53	76.7%	21.5379	2
GARCH	7.58	12.7955	+12.7923	+6.54	36.4%	21.6686	6
Heston	6.79	11.8187	+11.8187	+6.07	89.7%	30.9317	5
Merton	6.14	10.2893	+10.2846	+5.27	61.7%	21.4296	3
GARCH-Merton	7.62	12.8551	+12.8530	+6.57	42.7%	22.7237	7
Heston-Merton	6.50	11.3011	+11.3011	+5.80	89.7%	30.5776	4

Table SPY-3 SPY · 2018-2019 Late-cycle
2018-10-01 → 2019-09-30 · n = 251 days

Model	RMSE%	MAE	Bias \$	Bias%	ICP 25-75	ABW	Mark
GBM	11.83	31.0558	+31.0558	+11.16	1.2%	31.1821	2
Modified GBM	7.97	17.9884	+17.9884	+6.60	46.6%	29.2245	1
GARCH	16.38	43.8185	+43.8185	+15.60	1.2%	28.7700	5
Heston	13.51	35.8610	+35.8610	+12.85	2.4%	38.0925	4
Merton	18.97	50.7657	+50.7657	+18.03	1.2%	32.5920	7
GARCH-Merton	16.51	44.1759	+44.1759	+15.73	1.2%	32.8409	6
Heston-Merton	12.48	32.8911	+32.8911	+11.81	1.2%	30.6092	3

Table SPY-4 SPY · 2019-2020 COVID
2019-09-01 → 2020-08-31 · n = 252 days

Model	RMSE%	MAE	Bias \$	Bias%	ICP 25-75	ABW	Mark
GBM	8.54	20.6229	-9.5160	-2.50	49.2%	41.3009	3
Modified GBM	9.21	23.6787	-15.6556	-4.50	41.7%	41.5597	5
GARCH	12.31	25.0378	+18.0040	+6.48	61.9%	46.0917	6
Heston	8.43	17.9228	-2.3723	-0.15	67.9%	50.1048	2
Merton	8.58	18.2211	-1.2970	+0.18	59.1%	41.8094	4
GARCH-Merton	12.31	24.9973	+18.3131	+6.57	65.5%	49.7937	7
Heston-Merton	8.34	17.8798	-3.1895	-0.42	70.6%	54.2445	1

RMSE% is $100 \times \sqrt{\text{mean}(((p50 - S_t)/S_t)^2)}$. MAE and Bias \$ are in stock-price dollars. ICP 25-75 = share of S_t inside [p25, p75]. ABW = mean(p75-p25). Ranking uses RMSE% only.

Page 4 · Table block · AAPL · seven models × four regimes

P-measure daily cloud, reported path = p50. Bold green = lowest RMSE% vs realized S_t. 1.5-year lookback, rolling=none, n_paths=10000, seed=42.

Table AAPL-1 AAPL · 2008-2009 Crisis
2008-08-01 → 2009-07-31 · n = 252 days

Model	RMSE%	MAE	Bias \$	Bias%	ICP 25-75	ABW	Mark
GBM	73.54	2.1069	+2.0402	+64.72	4.8%	2.1188	2
Modified GBM	71.03	2.0176	+1.9533	+62.24	5.2%	1.9370	1
GARCH	110.72	3.3430	+3.2906	+100.24	5.2%	2.6794	7
Heston	77.34	2.2246	+2.1665	+68.47	4.4%	2.2178	4
Merton	79.75	2.3207	+2.2578	+70.92	4.4%	2.1865	5
GARCH-Merton	110.41	3.3318	+3.2803	+99.97	5.2%	2.7863	6
Heston-Merton	76.47	2.1943	+2.1348	+67.57	4.4%	2.1744	3

Table AAPL-2 AAPL · 2013-2014 Normal
2014-01-01 → 2014-12-31 · n = 253 days

Model	RMSE%	MAE	Bias \$	Bias%	ICP 25-75	ABW	Mark
GBM	21.23	3.9335	-3.6702	-15.85	32.8%	4.3434	4
Modified GBM	27.69	5.1179	-4.9949	-21.93	27.3%	3.7517	7
GARCH	12.66	2.4208	-1.8198	-7.37	61.7%	4.6797	1
Heston	21.86	4.0476	-3.7819	-16.34	33.6%	4.8853	5
Merton	14.31	2.7277	-2.1412	-8.79	52.6%	4.7267	3
GARCH-Merton	13.31	2.5465	-1.9428	-7.91	72.7%	5.4147	2
Heston-Merton	21.94	4.0571	-3.7984	-16.41	33.2%	4.7240	6

Table AAPL-3 AAPL · 2018-2019 Late-cycle
2018-10-01 → 2019-09-30 · n = 251 days

Model	RMSE%	MAE	Bias \$	Bias%	ICP 25-75	ABW	Mark
GBM	40.68	17.0222	+17.0107	+37.95	1.6%	12.1123	2
Modified GBM	43.92	18.5121	+18.5009	+41.11	1.6%	12.5894	7
GARCH	43.82	18.4671	+18.4559	+41.01	1.6%	12.3393	6
Heston	42.45	17.7797	+17.7690	+39.67	1.6%	19.7651	4
Merton	34.13	13.9045	+13.8915	+31.33	2.0%	11.2247	1
GARCH-Merton	43.62	18.3666	+18.3554	+40.81	1.6%	13.4838	5
Heston-Merton	41.36	17.2716	+17.2600	+38.57	1.2%	17.8788	3

Table AAPL-4 AAPL · 2019-2020 COVID
2019-09-01 → 2020-08-31 · n = 252 days

Model	RMSE%	MAE	Bias \$	Bias%	ICP 25-75	ABW	Mark
GBM	30.37	22.4802	-22.4802	-27.34	4.0%	13.7604	6
Modified GBM	32.61	24.1778	-24.1778	-29.46	1.2%	13.6055	7
GARCH	14.16	9.8430	-8.6211	-10.04	50.8%	20.4755	1
Heston	28.97	21.3178	-21.3178	-25.75	7.1%	15.1999	3
Merton	29.38	21.7077	-21.7077	-26.35	5.6%	13.7630	5
GARCH-Merton	14.45	10.0988	-8.9957	-10.46	54.0%	22.3912	2
Heston-Merton	29.35	21.5969	-21.5969	-26.10	6.7%	14.9476	4

RMSE% is $100 \times \sqrt{\text{mean}(((p50 - S_t)/S_t)^2)}$. MAE and Bias \$ are in stock-price dollars. ICP 25-75 = share of S_t inside [p25, p75]. ABW = mean(p75-p25). Ranking uses RMSE% only.

Page 5 · Table block · MSFT · seven models × four regimes

P-measure daily cloud, reported path = p50. Bold green = lowest RMSE% vs realized S_t. 1.5-year lookback, rolling=none, n_paths=10000, seed=42.

Table MSFT-1 MSFT · 2008-2009 Crisis
2008-08-01 → 2009-07-31 · n = 252 days

Model	RMSE%	MAE	Bias \$	Bias%	ICP 25-75	ABW	Mark
GBM	19.26	2.2887	+1.4334	+11.22	42.1%	4.1984	1
Modified GBM	22.72	2.5827	+2.0905	+15.54	34.5%	4.4448	7
GARCH	22.13	2.5272	+1.9849	+14.85	40.9%	5.0210	5
Heston	21.24	2.4490	+1.8161	+13.74	30.6%	3.1463	3
Merton	22.57	2.5699	+2.0563	+15.32	34.9%	4.3125	6
GARCH-Merton	21.83	2.5049	+1.9408	+14.55	44.8%	5.3805	4
Heston-Merton	20.71	2.4027	+1.7089	+13.04	33.7%	3.3700	2

Table MSFT-2 MSFT · 2013-2014 Normal
2014-01-01 → 2014-12-31 · n = 253 days

Model	RMSE%	MAE	Bias \$	Bias%	ICP 25-75	ABW	Mark
GBM	9.16	2.9735	-2.7770	-7.11	74.3%	7.1367	4
Modified GBM	13.58	4.4580	-4.2900	-11.07	41.1%	6.7242	7
GARCH	2.75	0.8224	+0.1736	+0.63	93.7%	8.1055	1
Heston	9.53	3.0966	-2.9099	-7.46	77.5%	7.4613	6
Merton	6.12	1.9212	-1.6448	-4.11	94.1%	7.2214	3
GARCH-Merton	2.86	0.8682	-0.0403	+0.09	95.3%	9.2508	2
Heston-Merton	9.24	2.9697	-2.7616	-7.04	75.9%	7.4106	5

Table MSFT-3 MSFT · 2018-2019 Late-cycle
2018-10-01 → 2019-09-30 · n = 251 days

Model	RMSE%	MAE	Bias \$	Bias%	ICP 25-75	ABW	Mark
GBM	16.53	17.7039	+17.7039	+15.66	15.1%	24.6968	1
Modified GBM	18.89	20.5816	+20.5816	+18.03	1.2%	23.5056	5
GARCH	23.93	26.4668	+26.4668	+22.87	1.2%	25.3303	7
Heston	17.26	18.4667	+18.4667	+16.37	43.0%	32.4861	3
Merton	18.64	20.2820	+20.2820	+17.80	2.4%	24.6924	4
GARCH-Merton	23.41	25.8543	+25.8543	+22.37	1.2%	28.2292	6
Heston-Merton	17.08	18.1738	+18.1738	+16.14	44.2%	33.2598	2

Table MSFT-4 MSFT · 2019-2020 COVID
2019-09-01 → 2020-08-31 · n = 252 days

Model	RMSE%	MAE	Bias \$	Bias%	ICP 25-75	ABW	Mark
GBM	12.14	17.8924	-16.6883	-9.02	64.3%	34.7272	6
Modified GBM	9.44	13.6285	-11.5096	-6.06	78.2%	33.4464	3
GARCH	6.81	8.8545	-5.0987	-2.42	90.9%	37.3130	1
Heston	10.59	15.3221	-13.2009	-6.93	73.4%	37.6172	4
Merton	14.58	21.6196	-20.9360	-11.45	47.6%	33.2491	7
GARCH-Merton	6.98	9.1410	-5.4056	-2.57	91.7%	40.2136	2
Heston-Merton	11.09	16.1049	-14.1374	-7.46	67.9%	35.7307	5

RMSE% is $100 \times \sqrt{\text{mean}(((p50 - S_t)/S_t)^2)}$. MAE and Bias \$ are in stock-price dollars. ICP 25-75 = share of S_t inside [p25, p75]. ABW = mean(p75-p25). Ranking uses RMSE% only.

Page 6 · Table block · AMZN · seven models × four regimes

P-measure daily cloud, reported path = p50. Bold green = lowest RMSE% vs realized S_t. 1.5-year lookback, rolling=none, n_paths=10000, seed=42.

Table AMZN-1 AMZN · 2008-2009 Crisis
2008-08-01 → 2009-07-31 · n = 252 days

Model	RMSE%	MAE	Bias \$	Bias%	ICP 25-75	ABW	Mark
GBM	46.90	1.1152	+1.0618	+36.69	49.6%	2.1141	2
Modified GBM	114.01	3.4021	+3.3747	+102.32	10.3%	3.1462	7
GARCH	61.04	1.6451	+1.6010	+52.15	12.7%	2.4144	6
Heston	52.88	1.3088	+1.2678	+42.93	44.8%	2.1380	4
Merton	29.61	0.6553	+0.1297	+9.51	61.1%	1.6009	1
GARCH-Merton	56.44	1.4796	+1.4290	+47.23	37.7%	2.7035	5
Heston-Merton	49.13	1.1873	+1.1397	+39.05	40.9%	1.9112	3

Table AMZN-2 AMZN · 2013-2014 Normal
2014-01-01 → 2014-12-31 · n = 253 days

Model	RMSE%	MAE	Bias \$	Bias%	ICP 25-75	ABW	Mark
GBM	49.05	6.9955	+6.9931	+43.67	6.7%	5.8701	4
Modified GBM	55.85	7.9140	+7.9130	+49.42	6.7%	6.1535	5
GARCH	61.42	8.6589	+8.6582	+54.08	5.9%	6.4431	7
Heston	42.80	6.2650	+6.2646	+39.04	43.5%	11.6254	2
Merton	39.35	5.6550	+5.6498	+35.30	7.1%	5.4556	1
GARCH-Merton	60.00	8.4580	+8.4567	+52.83	6.7%	7.2906	6
Heston-Merton	45.25	6.5505	+6.5491	+40.85	6.7%	9.8435	3

Table AMZN-3 AMZN · 2018-2019 Late-cycle
2018-10-01 → 2019-09-30 · n = 251 days

Model	RMSE%	MAE	Bias \$	Bias%	ICP 25-75	ABW	Mark
GBM	51.83	42.8456	+42.8456	+48.55	0.4%	29.8080	4
Modified GBM	62.88	51.4866	+51.4866	+58.13	0.4%	34.2508	5
GARCH	69.97	56.9143	+56.9143	+64.12	0.4%	33.8579	7
Heston	50.46	42.0118	+42.0118	+47.72	0.4%	54.5373	2
Merton	50.76	41.9602	+41.9602	+47.56	0.4%	28.3860	3
GARCH-Merton	68.07	55.4059	+55.4059	+62.44	0.4%	37.8848	6
Heston-Merton	49.84	41.5285	+41.5285	+47.18	0.4%	51.6194	1

Table AMZN-4 AMZN · 2019-2020 COVID
2019-09-01 → 2020-08-31 · n = 252 days

Model	RMSE%	MAE	Bias \$	Bias%	ICP 25-75	ABW	Mark
GBM	20.22	19.3730	-17.9219	-12.62	57.9%	27.3426	6
Modified GBM	18.09	17.3793	-14.8372	-9.99	63.5%	29.3614	4
GARCH	12.90	11.2788	+6.1144	+7.45	52.0%	33.6315	1
Heston	18.86	18.2177	-15.3033	-10.24	56.0%	25.5770	5
Merton	20.65	19.7910	-18.4425	-13.04	57.1%	26.8815	7
GARCH-Merton	12.97	11.3942	+6.0360	+7.42	52.4%	34.4570	2
Heston-Merton	17.12	17.2476	-10.9413	-6.29	70.6%	38.3923	3

RMSE% is $100 \times \sqrt{\text{mean}(((p50 - S_t)/S_t)^2)}$. MAE and Bias \$ are in stock-price dollars. ICP 25-75 = share of S_t inside [p25, p75]. ABW = mean(p75-p25). Ranking uses RMSE% only.

Page 7 · Summary tables · RMSE% across companies and regimes

Table S1 · Overall ranking (mean RMSE% over 16 cells)

Rank	Model	Mean RMSE%	Median RMSE%	# of 16 cells best
1	Merton	26.52	21.61	4
2	GBM	28.17	20.73	3
3	Heston-Merton	28.31	21.32	2
4	Heston	28.79	21.55	0
5	GARCH-Merton	31.91	19.17	0
6	GARCH	32.40	19.25	5
7	Modified GBM	34.32	25.21	2

Table S2 · Wins by company (lowest RMSE% in that company×regime)

Model	SPY wins	AAPL wins	MSFT wins	AMZN wins	Total
GBM	1	0	2	0	3
Modified GBM	1	1	0	0	2
GARCH	0	2	2	1	5
Heston	0	0	0	0	0
Merton	1	1	0	2	4
GARCH-Merton	0	0	0	0	0
Heston-Merton	1	0	0	1	2

Table S3 · Mean RMSE% by regime (equal-weight mean of SPY / AAPL / MSFT / AMZN)

Model	2008-2009 Crisis	2013-2014 Normal	2018-2019 Late-cycle	2019-2020 COVID	Mean of 4
GBM	43.53	21.11	30.22	17.82	28.17
Modified GBM	60.90	25.62	33.41	17.34	34.32
GARCH	58.44	21.10	38.52	11.55	32.40
Heston	47.30	20.25	30.92	16.71	28.79
Merton	40.69	16.48	30.63	18.30	26.52
GARCH-Merton	57.11	20.95	37.90	11.68	31.91
Heston-Merton	45.84	20.73	30.19	16.48	28.31

Green row in S1 / S3 = lowest mean RMSE%. Green row in S2 = most company×regime wins. A model can win the most cells without having the lowest average error (and vice versa).

Table S4.1 · SPY

Model	Mean RMSE%	Median	# regimes best
GBM	14.94	10.18	1
Modified GBM	14.58	8.59	1
GARCH	19.04	14.34	0
Heston	16.62	10.97	0
Merton	16.13	13.77	1
GARCH-Merton	19.05	14.41	0
Heston-Merton	16.10	10.41	1

Table S4.2 · AAPL

Model	Mean RMSE%	Median	# regimes best
GBM	41.45	35.52	0
Modified GBM	43.81	38.27	1
GARCH	45.34	28.99	2
Heston	42.66	35.71	0
Merton	39.39	31.76	1
GARCH-Merton	45.45	29.04	0
Heston-Merton	42.28	35.35	0

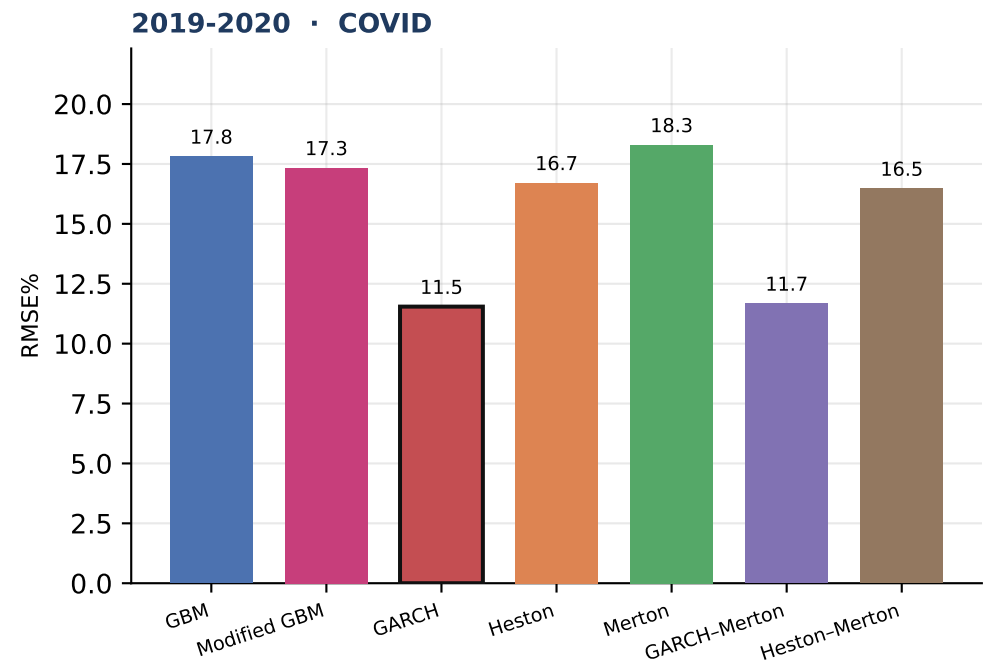
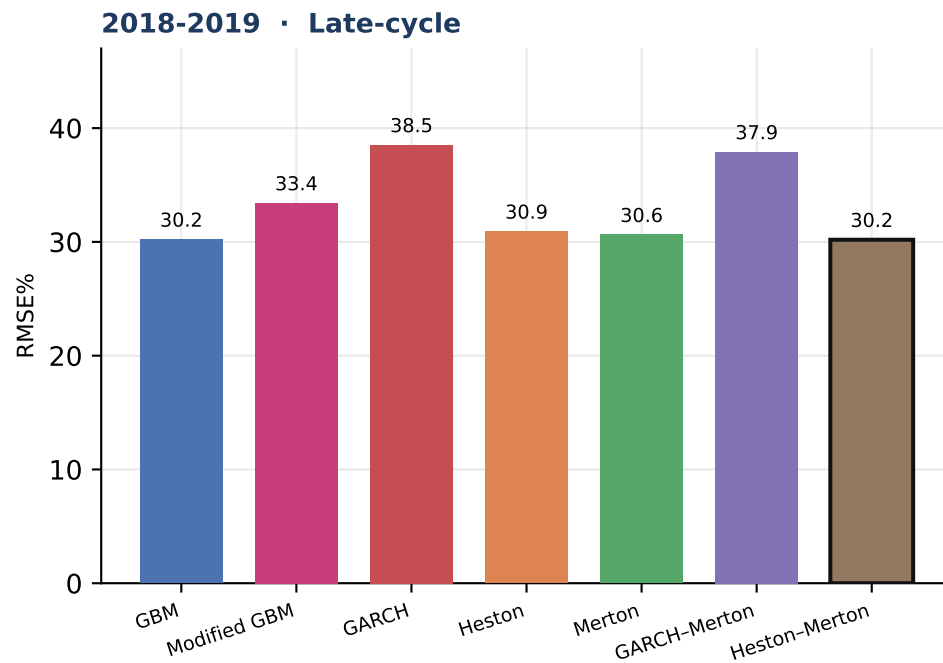
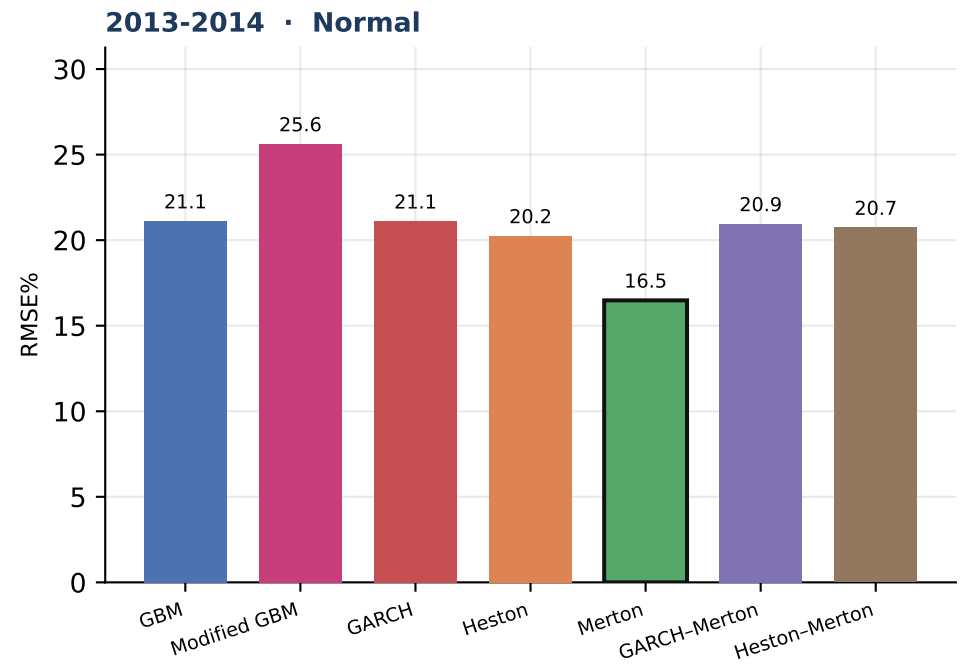
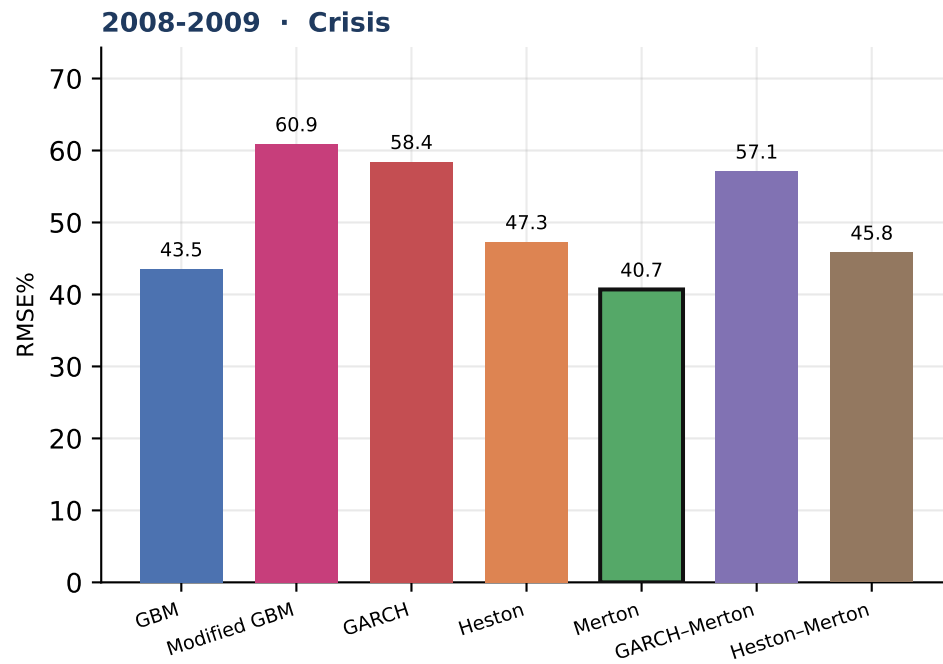
Table S4.3 · MSFT

Model	Mean RMSE%	Median	# regimes best
GBM	14.27	14.33	2
Modified GBM	16.16	16.23	0
GARCH	13.90	14.47	2
Heston	14.66	13.92	0
Merton	15.48	16.61	0
GARCH-Merton	13.77	14.40	0
Heston-Merton	14.53	14.09	0

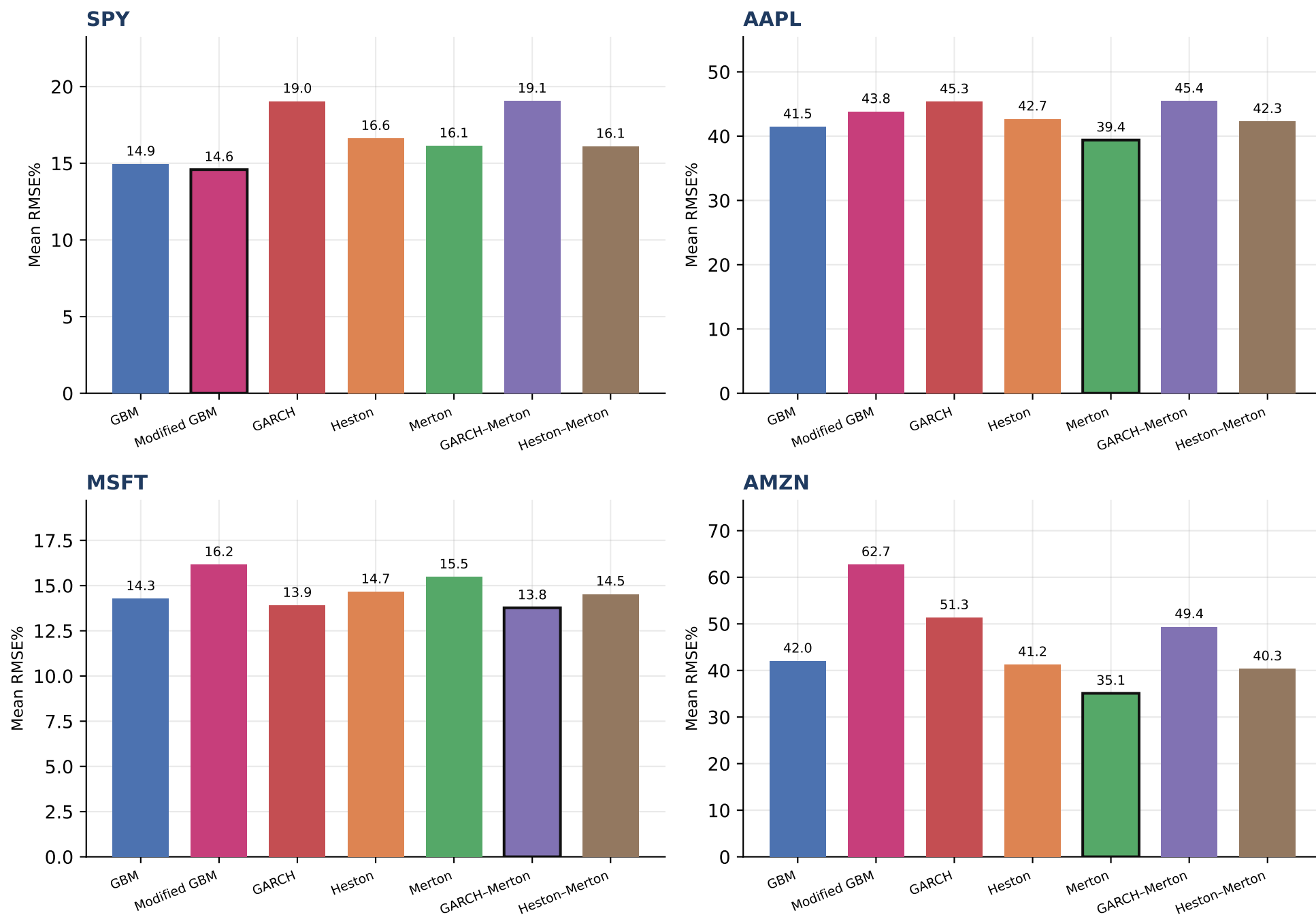
Table S4.4 · AMZN

Model	Mean RMSE%	Median	# regimes best
GBM	42.00	47.98	0
Modified GBM	62.71	59.36	0
GARCH	51.33	61.23	1
Heston	41.25	46.63	0
Merton	35.09	34.48	2
GARCH-Merton	49.37	58.22	0
Heston-Merton	40.34	47.19	1

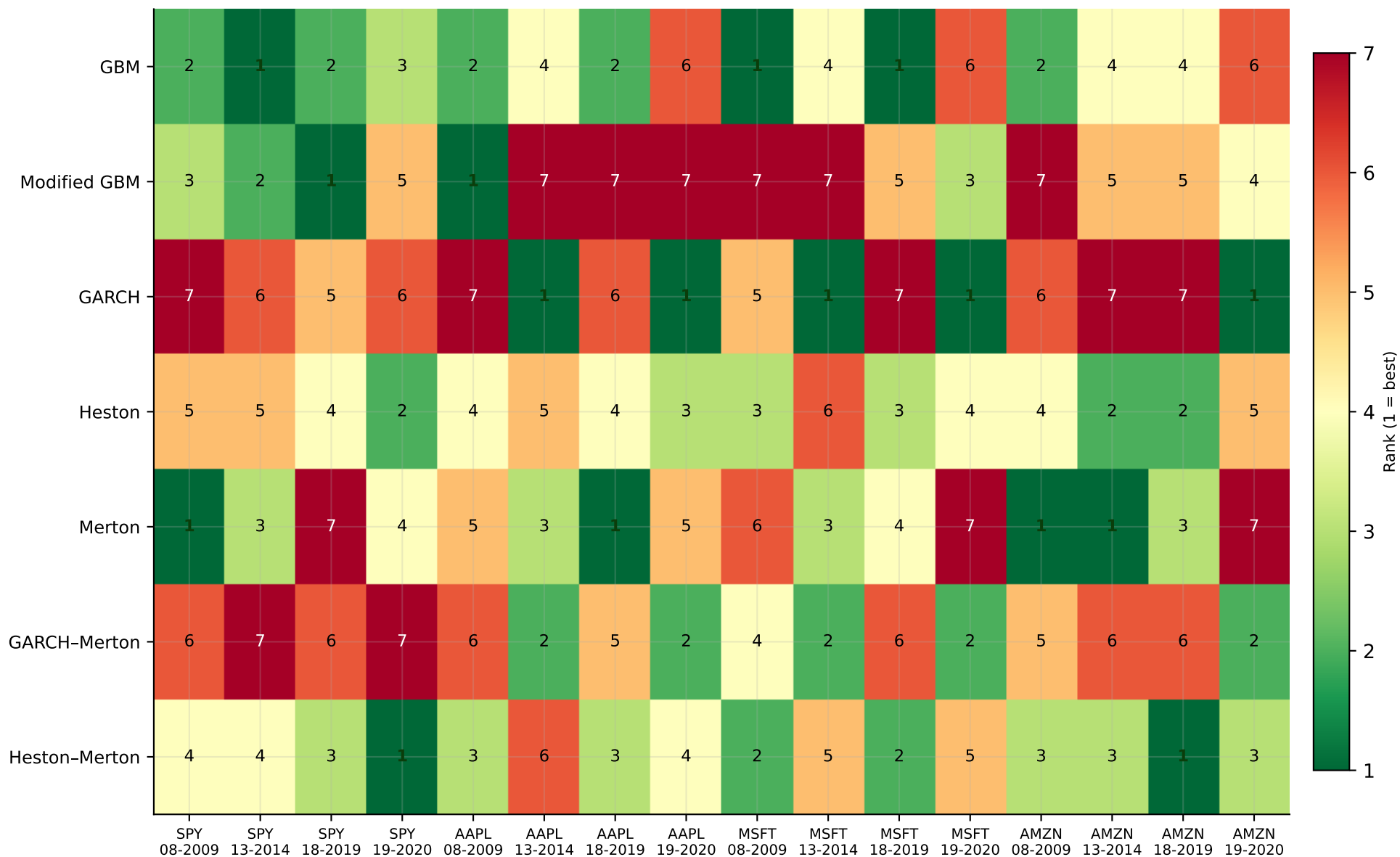
Each panel is one company. Green row = lowest mean RMSE% across the four V3 regimes for that name.



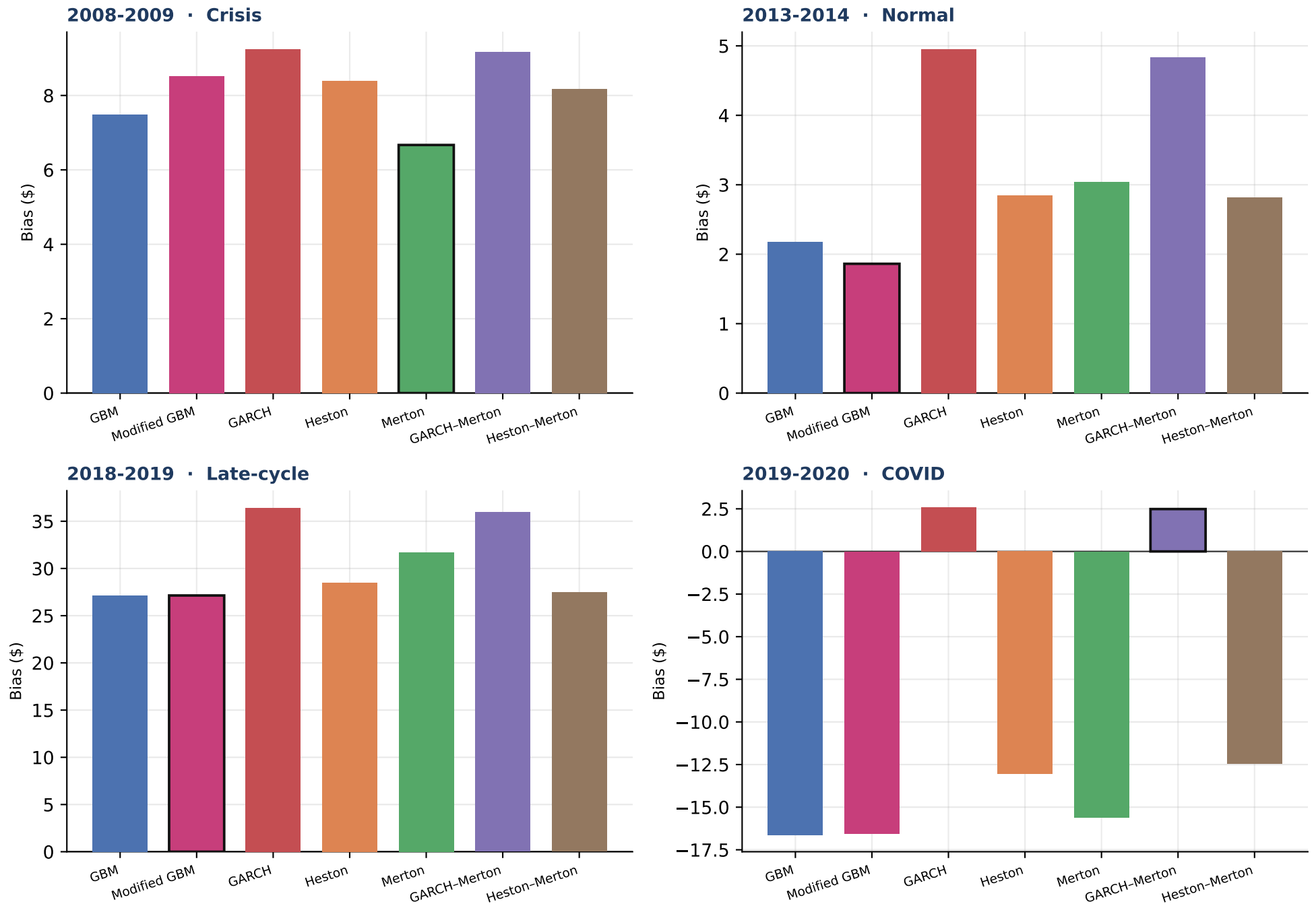
Outlined bar = lowest mean RMSE% in that regime. Equal-weight mean of the four companies.



Outlined bar = lowest mean RMSE% for that company.



Green / 1 = lowest RMSE% in that company×regime cell. Rankings can and do change across names and volatility states.



Outlined bar = closest to zero bias (not the RMSE ranking). Positive = model expensive vs listed quote.

Final analysis

Which models track realized S_t most closely?

- On mean RMSE% of the p50 path versus realized S_t across 16 company×regime cells, Merton is best (mean RMSE% = 26.52; 4 of 16 cells).
- The ranking is not stable. Best-in-cell models: GARCH, GBM, Heston–Merton, Merton, Modified GBM.
- These conclusions are conditional on 1.5-year fixed calibration P-measure calibration, 10000 paths, daily steps, and the p50 scoring convention.